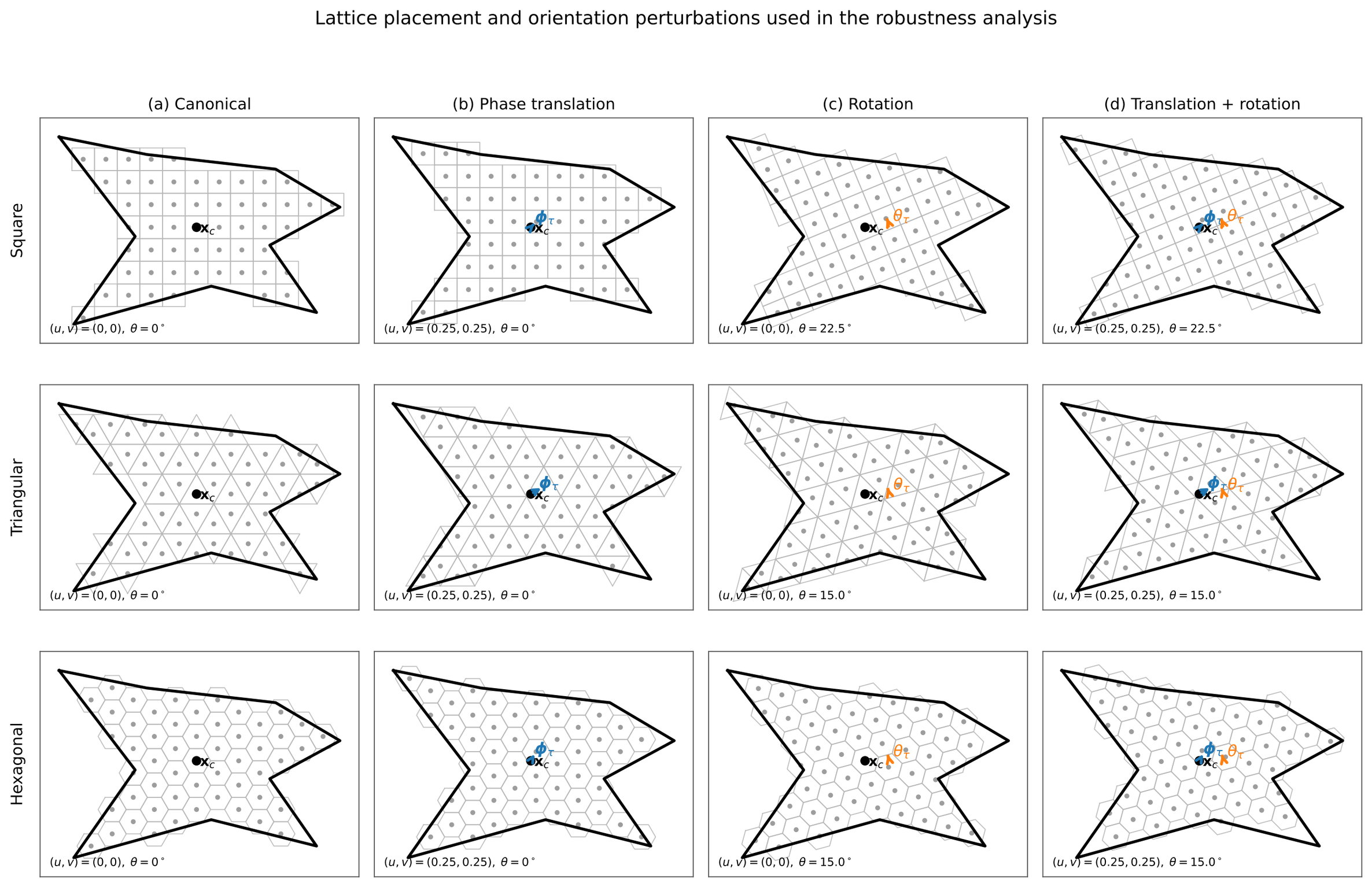

Lattice placement and orientation perturbations used in the robustness analysis
(a) Canonical
(b) Phase translation
(c) Rotation
(d) Translation + rotation
Square
Triangular
Hexagonal
$\mathbf{x}_c$
$\boldsymbol{\phi}_\tau$
$\theta_\tau$
$(u, v) = (0, 0),\ \theta = 0^\circ$
$(u, v) = (0.25, 0.25),\ \theta = 0^\circ$
$(u, v) = (0, 0),\ \theta = 22.5^\circ$
$(u, v) = (0.25, 0.25),\ \theta = 22.5^\circ$
$(u, v) = (0, 0),\ \theta = 0^\circ$
$(u, v) = (0.25, 0.25),\ \theta = 0^\circ$
$(u, v) = (0, 0),\ \theta = 15.0^\circ$
$(u, v) = (0.25, 0.25),\ \theta = 15.0^\circ$
$(u, v) = (0, 0),\ \theta = 0^\circ$
$(u, v) = (0.25, 0.25),\ \theta = 0^\circ$
$(u, v) = (0, 0),\ \theta = 15.0^\circ$
$(u, v) = (0.25, 0.25),\ \theta = 15.0^\circ$